

[Aim: 100/100 in Maths]

आरंभ

QUADRATIC EQUATIONS

Lecture #3

#ATK²:

Q.E $\rightarrow$ परीक्षण ? $\Rightarrow$ $D = 2$

Q.E के roots

$\rightarrow$ Splitting the M.T. / Factorisation ✓

$\rightarrow$ Discriminant method

#LP: $\frac{2x}{x-3} + \frac{1}{2x+3} + \frac{3x+9}{(x-3)(2x+3)} = 0, \boxed{x \neq 3}, \frac{-3}{2}$

$$\Rightarrow \frac{2x(2x+3) + 1(x-3) + (3x+9)}{(x-3)(2x+3)} = 0$$

$$\Rightarrow \frac{4x^2 + 6x + x - 3 + 3x + 9}{(x-3)(2x+3)} = 0$$

$$\Rightarrow \frac{4x^2 + 10x + 6}{(x-3)(2x+3)} = 0$$

$$\Rightarrow \frac{4x^2 + 6x + 4x + 6}{(x-3)(2x+3)} = 0$$

$$\Rightarrow \frac{2x(2x+3) + 1(2x+3)}{(x-3)(2x+3)} = 0$$

$$\Rightarrow \frac{(2x+3)[2x+1]}{(x-3)(2x+3)} = 0$$

$$\Rightarrow \frac{(2x+1)}{(x-3)} = 0$$

$$2x+1=0$$

$$2x = -1$$

$$\boxed{x = -\frac{1}{2}}$$

h

$$x-3 \neq 0$$

$$\boxed{x \neq 3}$$

#LP: Solve the quadratic equation: $x^2 - 2ax + (a^2 - b^2) = 0$ for x .
(CBSE 2022)

$$x^2 - 2ax + (a^2 - b^2) = 0$$

$$\Rightarrow x^2 - 2ax + (a+b)(a-b) = 0$$

$$\Rightarrow x^2 - (a+b)x - (a-b)x + (a+b)(a-b) = 0$$

$$\Rightarrow x[x - (a+b)] - (a-b)[x - (a+b)] = 0$$

$$\Rightarrow [x - (a+b)][x - (a-b)] = 0$$

$$\begin{array}{l|l} x - (a+b) = 0 & x - (a-b) = 0 \\ \hline x = (a+b) \checkmark & x = (a-b) \checkmark \end{array}$$

$$\begin{aligned} p+q &= -2a \\ pq &= (a+b)(a-b) \end{aligned}$$

$$a^2 - b^2 \Rightarrow (a+b)(a-b)$$

$$\begin{array}{cc} p & q \\ - (a+b) & - (a-b) \end{array}$$

$$\Rightarrow -a - b - a + b$$

$$\Rightarrow -2a$$

$$\#LP: x^2 - 2x - (r^2 - 1) = 0$$

$$\Rightarrow x^2 - 2x - (r+1)(r-1) = 0$$

$$\Rightarrow \underbrace{x^2 - (r+1)x}_{p} + \underbrace{(r-1)x - (r+1)(r-1)}_{q} = 0$$

$$\Rightarrow \underline{x} \left[\underline{x - (r+1)} \right] + \underline{(r-1)} \left[\underline{x - (r+1)} \right] = 0$$

$$\Rightarrow \left[x - (r+1) \right] \left[x + (r-1) \right] = 0$$

$$x - (r+1) = 0 \quad | \quad x + (r-1) = 0$$

$$\boxed{x = (r+1)} \quad | \quad \boxed{x = -(r-1)}$$

$$p+q = -2 \checkmark$$

$$pq = -(r+1)(r-1)$$

$$\begin{aligned} r^2 - 1 &\Rightarrow r^2 - 1^2 \\ &\Rightarrow (r+1)(r-1) \end{aligned}$$

$$\begin{aligned} &\underbrace{-(r+1)}_p + \underbrace{(r-1)}_q \\ &\quad -r-1 + r-1 \\ &\quad \Rightarrow -2 \end{aligned}$$

#LP: $9x^2 - 6b^2x - (a^4 - b^4) = 0$ (CBSE 2015)

$$\Rightarrow 9x^2 - 6b^2x - (a^2+b^2)(a^2-b^2) = 0$$

$$\begin{array}{r|l} 3 & 9 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$\Rightarrow 9x^2 - 3(a^2+b^2)x + 3(a^2-b^2)x - (a^2+b^2)(a^2-b^2) = 0$$

$$\Rightarrow 3x[3x - (a^2+b^2)] + (a^2-b^2)[3x - (a^2+b^2)] = 0$$

$$\Rightarrow [3x - (a^2+b^2)][3x + (a^2-b^2)] = 0$$

$$\Rightarrow 3x - (a^2+b^2) = 0 \quad \vee \quad 3x + (a^2-b^2) = 0$$

$$\Rightarrow 3x = (a^2+b^2)$$

$$x = \frac{1}{3}(a^2+b^2)$$

$$\Rightarrow 3x = -(a^2-b^2)$$

$$x = -\frac{1}{3}(a^2-b^2)$$

✓

$$\frac{a^4 - b^4}{(a^2+b^2)(a^2-b^2)} = \frac{(a^2)^2 - (b^2)^2}{(a^2+b^2)(a^2-b^2)}$$

$$p+q = -6b^2$$

$$pq = -9(a^2+b^2)(a^2-b^2)$$

$$3 \cdot 3(a^2+b^2)(a^2-b^2)$$

$$-3(a^2+b^2) + 3(a^2-b^2)$$

$$\Rightarrow -3a^2 - 3b^2 + 3a^2 - 3b^2$$

$$\Rightarrow -6b^2$$

#LP: Solve the quadratic equation by factorization method:

$$9x^2 - 9(a+b)x + (2a^2 + 5ab + 2b^2) = 0 \quad (\text{CBSE 2004, 2009, 2016})$$

$$\Rightarrow 9x^2 - 9(a+b)x + (a+2b)(2a+b) = 0$$

$$\Rightarrow 9x^2 - 3(a+2b)x - 3(2a+b)x + (a+2b)(2a+b) = 0$$

$$\Rightarrow 3x[3x - (a+2b)] - (2a+b)[3x - (a+2b)] = 0$$

$$\Rightarrow [3x - (a+2b)][3x - (2a+b)] = 0$$

$$\Rightarrow 3x - (a+2b) = 0$$

$$\Rightarrow 3x = (a+2b)$$

$$\Rightarrow x = \frac{a+2b}{3} \checkmark$$

$$\Rightarrow 3x - (2a+b) = 0$$

$$\Rightarrow 3x = (2a+b)$$

$$\Rightarrow x = \frac{2a+b}{3} \checkmark$$

$$2a^2 + 5ab + 2b^2$$

$$\Rightarrow 2a^2 + 4ab + 1ab + 2b^2$$

$$\Rightarrow 2a(a+2b) + b(a+2b)$$

$$(a+2b)(2a+b)$$

$$p+q = -9(a+b)$$

$$pq = 9(a+2b)(2a+b)$$

$$-3(a+2b) \quad -3(2a+b)$$

$$- 3(a+2b) - 3(2a+b)$$

$$\Rightarrow \underline{-3a} - \underline{6b} - \underline{6a} - \underline{3b}$$

$$\uparrow \quad -9a - 9b$$

$$\uparrow \boxed{-9(a+b)} \checkmark$$

$$\begin{array}{r} b+a \\ \hline -9(a+b) \end{array}$$